Theory track

Each chapter follows the same structure:

1. **Motivation** — why does this matter?
2. **Definitions** — precise statements of objects we'll manipulate.
3. **Derivations / theorems** — the math, with proofs where short enough to be illuminating.
4. **Worked example** — a tiny numerical case you can verify by hand.
5. **Exercises** — numbered, with difficulty markers:
   - ⋆ — routine, should take < 5 min
   - ⋆⋆ — requires real thought, 10–30 min
   - ⋆⋆⋆ — open-ended or a small proof, 30+ min
6. **Before the notebook** — a prediction you must write down before running code.

Do exercises in a paper notebook or a scratch markdown file. Solutions are in `theory/solutions/` — resist until you've made a genuine attempt.

Math prerequisites

- Linear algebra: matrix multiplication, inner products, eigendecomposition, norms.
- Probability: discrete distributions, expectation, variance, independence.
- Calculus: partial derivatives, chain rule, gradient of scalar w.r.t. vector.

If any of those feel shaky, start with `00_linear_algebra_primer.md`.